

COMP2100 Hackathon: prediction index

Keep this open during the hackathon. Built from the practice hackathon + your miniproject. Every entry points at tested code in the **Predictions** folder (201 tests pass; 39/39 broken versions caught).

How to use it (30 seconds per question)

- 1 Read the question and pick out its nouns and verbs ("report", "hide", "oldest", "vote", "block"...).
- 2 Find those words in the **Keyword index** (pages 2-6). It gives the theme number and where its code is.
- 3 Open that file at that section. **T1 s5** means **Task1prediction.md**, section 5. Search the file for "## 5." to jump there.
- 4 No match? Use the **Task shapes** table below: the five tasks always ask for the same kind of code, whatever the theme.

File codes

Code	File (in the Predictions folder)	What is in it
T1	Task1prediction.md	core features: reports, reactions, votes, follow/block/bookmark, notifications, tags, polls, read tracking, bans, friends, mentions, edits, accounts, rate limit; s3 = base fixes
T2	Task2prediction.md	Post state (hide, delete, pin, lock), ContentTools, AVL delete, rank/floor/ceiling, ranges, pages, BST iterator, red-black tree, move
T3	Task3prediction.md	persistence: s2 = two base bugs that break ALL loading; s6 = every feature saved; s7 = JSON
T4	Task4prediction.md	Iterator + Factory queries; s4 = reusable TopKIterator
T5	Task5prediction.md	white-box + black-box tests, branch maps, mutation table
Extra	ExtraPredictions.md	search, stats, censor, audit/undo, DMs, scheduled, boards, links, replies/threads, private posts, login lockout, merge/split, emoji/markdown
UML	UMLprediction.md, uml.png, uml_reactions.png	the group UML task
zip	verified-code.zip	the whole working project: open the full file when a copied snippet will not compile

Task shapes (what each task always asks for)

Task	Shape	Marks	Start here
Task 1	core feature: add / remove / has	50% behaviour, 30% efficiency, 20% quality	Task1prediction.md s3 (indexes) + the theme's section
Task 2	state on a model class + a SortedData view, 'trees'	40 / 40 / 20	Task2prediction.md s3-s5 (+ s7 AVL delete)
Task 3	persistence + refactoring, 'portable'	60% persistence, 40% code smells	Task3prediction.md s2 (two base bugs!), s6 (every feature)
Task 4	query with a strategy string, Iterator + Factory	40 / 40 / 20	Task4prediction.md s3-s4 (TopKIterator)
Task 5	white-box branch coverage + black-box vs faulty versions	40 / 40 / 20	Task5prediction.md s2-s3 (+ branch maps)
UML	5 classes, - # + fields, 2 static + 2 non-static, composition / aggregation / association	50 / 50	UMLprediction.md + uml.png

First 15 minutes (do these before splitting up)

- [] Everyone reads ALL tasks; agree on the data model (store classes, maps, where new state lives, file format).
- [] Base fixes, then push: **MessageIndex** + UUID index in **UserDAO** (T1 s3), null-safe **MessageComparator** (T1 s3.3), AVL in **SortedDataFactory** (T2 s7).
- [] If the base reads/writes files: fix **CSVReader.hasNext** and remove the **ComputerIOFactory** date check (T3 s2). Otherwise nothing ever loads.
- [] Final commit on **main** must COMPILE. Compile + run tests before the last push. Last 20 min: javadoc, remove dead code, uml.png.

Keyword index

Words a question might use, A-Z. "#" is the theme number (full list on pages 7-8). Red = deliberately not coded.

Keyword	#	Theme	Code
#	21	Tags / hashtags	T1 s9
abusive	3	Reports / moderation (the practice)	T1 s4; T4 s3, s5; T5 s2, s3
accept	13	Friend requests	T1 s13
access	15	Private / invite-only posts	Extra s10
alert	12b	Subscribe + notifications	T1 s8
alert	36	Notifications inbox	T1 s8
analytics	26	Statistics	Extra s2
archive	29	Lock / close a post	T2 s5, s6
attachment	41	Links / emoji / markdown / previews	Extra s8, s13
attempts	20	Login logout + login history	Extra s11
audit	40	Audit log	Extra s4
autocomplete	23	Search + autocomplete	Extra s1
average	26	Statistics	Extra s2
B-tree	F8	Skip list / heap / hash table / B-tree / Fenwick tree	not coded: T2 s11 (copy RedBlackTree's shape)
backwards	F3	SortedList backwards + bounds	T2 s7
balanced tree	F4	Red-black tree	T2 s11
ballot	10	Polls	T1 s10
ban	17	Bans with expiry	T1 s12
banned words	33	Profanity censor / word filter	Extra s3
between	32	Time-range queries / counts	T2 s9
binary search tree iterator	F2	BSTree.getRange (was null)	T2 s10
block	11	Block / mute users	T1 s6
board	22	Boards / categories / subforums	Extra s7
bookmark	9	Bookmarks / favourites	T1 s6
BSTree	F2	BSTree.getRange (was null)	T2 s10
busiest	24	Trending / busiest / active posts	T4 s7
category	22	Boards / categories / subforums	Extra s7
category label	21	Tags / hashtags	T1 s9
ceiling	F6	floor / ceiling / successor / predecessor	T2 s8
censor	33	Profanity censor / word filter	Extra s3
change password	19	Change password / username	T1 s16
change username	19	Change password / username	T1 s16
chat between two users	14	Direct / private messages	Extra s5
child	6	Replies to a message, threads, quoting	Extra s9
choice	10	Polls	T1 s10
close	29	Lock / close a post	T2 s5, s6
closed	29	Lock / close a post	T2 s5, s6
combine posts	30	Move message / merge / split posts	T2 s12; Extra s12
command	38	Undo / redo admin actions	Extra s4
conversation	14	Direct / private messages	Extra s5
count in period	32	Time-range queries / counts	T2 s9
decline	13	Friend requests	T1 s13
delayed	28	Scheduled / delayed messages	Extra s6
delete	4	Hide / soft delete / restore	T2 s4, s5
delete account	18	Account deletion	T1 s16
delete from tree	F1	AVL delete	T2 s7
[deleted]	18	Account deletion	T1 s16
deletion	F1	AVL delete	T2 s7
demote	16	Moderator role, roles	T1 s16
direct message	14	Direct / private messages	Extra s5
DM	14	Direct / private messages	Extra s5
does not exist	39	O(1) lookup index	T1 s3

Keyword	#	Theme	Code
downvote	2	Up/down votes, score	T1 s7; T4 s6; T5 s6, s9
draft	28	Scheduled / delayed messages	Extra s6
duplicate	35	Duplicate-message detection	Extra s8
edit	5	Edit + edit history (+ undo)	T1 s15
efficient	39	O(1) lookup index	T1 s3
emoji	41	Links / emoji / markdown / previews	Extra s8, s13
emoji reaction	1	Reactions	T1 s5; T4 s4; T5 s5, s8
export	37	Export / import, JSON	T3 s6, s7
failed login	20	Login lockout + login history	Extra s11
favourite	9	Bookmarks / favourites	T1 s6
featured	8	Pin messages	T2 s5
feed	12	Follow users	T1 s6; T5 s7
filter	F7	Filtered view of a SortedData	T2 s3
filter words	33	Profanity censor / word filter	Extra s3
find	23	Search + autocomplete	Extra s1
flag	3	Reports / moderation (the practice)	T1 s4; T4 s3, s5; T5 s2, s3
flood	34	Spam / rate limiting	T1 s17
floor	F6	floor / ceiling / successor / predecessor	T2 s8
follow	12	Follow users	T1 s6; T5 s7
follower	12	Follow users	T1 s6; T5 s7
following	12	Follow users	T1 s6; T5 s7
friend	13	Friend requests	T1 s13
friend request	13	Friend requests	T1 s13
from ... to	32	Time-range queries / counts	T2 s9
GDPR	18	Account deletion	T1 s16
hash table	F8	Skip list / heap / hash table / B-tree / Fenwick tree	not coded: T2 s11 (copy RedBlackTree's shape)
hashtag	21	Tags / hashtags	T1 s9
heap	F8	Skip list / heap / hash table / B-tree / Fenwick tree	not coded: T2 s11 (copy RedBlackTree's shape)
hidden	4	Hide / soft delete / restore	T2 s4, s5
hide	4	Hide / soft delete / restore	T2 s4, s5
hide user	11	Block / mute users	T1 s6
hierarchy	22	Boards / categories / subforums	Extra s7
history	5	Edit + edit history (+ undo)	T1 s15
hot	24	Trending / busiest / active posts	T4 s7
how many before	F5	Order statistics	T2 s8, s9
ignore	11	Block / mute users	T1 s6
import	37	Export / import, JSON	T3 s6, s7
inbox	12b	Subscribe + notifications	T1 s8
inbox	36	Notifications inbox	T1 s8
index of	F5	Order statistics	T2 s8, s9
invite	15	Private / invite-only posts	Extra s10
JSON	37	Export / import, JSON	T3 s6, s7
k-th	F5	Order statistics	T2 s8, s9
karma	2	Up/down votes, score	T1 s7; T4 s6; T5 s6, s9
karma	25	Leaderboards / karma	T4 s8
keyword	23	Search + autocomplete	Extra s1
label	21	Tags / hashtags	T1 s9
last read	27	Read / unread tracking	T1 s11
later	28	Scheduled / delayed messages	Extra s6
leaderboard	25	Leaderboards / karma	T4 s8
like/love/laugh	1	Reactions	T1 s5; T4 s4; T5 s5, s8
link	41	Links / emoji / markdown / previews	Extra s8, s13
lock	29	Lock / close a post	T2 s5, s6
locked	29	Lock / close a post	T2 s5, s6
lockout	20	Login lockout + login history	Extra s11

Keyword	#	Theme	Code
log	40	Audit log	Extra s4
login history	20	Login logout + login history	Extra s11
lookup by id	39	O(1) lookup index	T1 s3
markdown	41	Links / emoji / markdown / previews	Extra s8, s13
mention	7	@mentions	T1 s14
merge	30	Move message / merge / split posts	T2 s12; Extra s12
moderate	3	Reports / moderation (the practice)	T1 s4; T4 s3, s5; T5 s2, s3
moderator	16	Moderator role, roles	T1 s16
moderator review	3	Reports / moderation (the practice)	T1 s4; T4 s3, s5; T5 s2, s3
modify message	5	Edit + edit history (+ undo)	T1 s15
most active user	25	Leaderboards / karma	T4 s8
move	30	Move message / merge / split posts	T2 s12; Extra s12
mute	11	Block / mute users	T1 s6
nearest	F6	floor / ceiling / successor / predecessor	T2 s8
nested	6	Replies to a message, threads, quoting	Extra s9
new since	27	Read / unread tracking	T1 s11
new SortedData	F4	Red-black tree	T2 s11
next	F6	floor / ceiling / successor / predecessor	T2 s8
notification	12b	Subscribe + notifications	T1 s8
notification	36	Notifications inbox	T1 s8
notify	12b	Subscribe + notifications	T1 s8
option	10	Polls	T1 s10
other languages	37	Export / import, JSON	T3 s6, s7
page	31	Pagination	T2 s9
page size	31	Pagination	T2 s9
paginate	31	Pagination	T2 s9
parent	6	Replies to a message, threads, quoting	Extra s9
per day	26	Statistics	Extra s2
per minute	34	Spam / rate limiting	T1 s17
per user	26	Statistics	Extra s2
performance	39	O(1) lookup index	T1 s3
permission	16	Moderator role, roles	T1 s16
permission to read	15	Private / invite-only posts	Extra s10
pin	8	Pin messages	T2 s5
pinned	8	Pin messages	T2 s5
pinned	F9	SortedDataSlice / AVLTreeSlice	not coded: week 6 lab (hidden implementation)
poll	10	Polls	T1 s10
popular	24	Trending / busiest / active posts	T4 s7
portable	37	Export / import, JSON	T3 s6, s7
position	F5	Order statistics	T2 s8, s9
prefix	23	Search + autocomplete	Extra s1
preview	41	Links / emoji / markdown / previews	Extra s8, s13
previous	F6	floor / ceiling / successor / predecessor	T2 s8
previous version	5	Edit + edit history (+ undo)	T1 s15
private message	14	Direct / private messages	Extra s5
private post	15	Private / invite-only posts	Extra s10
profanity	33	Profanity censor / word filter	Extra s3
promote	16	Moderator role, roles	T1 s16
publish at	28	Scheduled / delayed messages	Extra s6
query	23	Search + autocomplete	Extra s1
quote	6	Replies to a message, threads, quoting	Extra s9
range	32	Time-range queries / counts	T2 s9
rank	F5	Order statistics	T2 s8, s9
rate limit	34	Spam / rate limiting	T1 s17
rating	2	Up/down votes, score	T1 s7; T4 s6; T5 s6, s9

Keyword	#	Theme	Code
react	1	Reactions	T1 s5; T4 s4; T5 s5, s8
reaction	1	Reactions	T1 s5; T4 s4; T5 s5, s8
ReactionType	1	Reactions	T1 s5; T4 s4; T5 s5, s8
read	27	Read / unread tracking	T1 s11
read-only	29	Lock / close a post	T2 s5, s6
rebalance	F1	AVL delete	T2 s7
recent activity	24	Trending / busiest / active posts	T4 s7
record of admin actions	40	Audit log	Extra s4
red-black	F4	Red-black tree	T2 s11
redo	38	Undo / redo admin actions	Extra s4
remove	F1	AVL delete	T2 s7
remove message	4	Hide / soft delete / restore	T2 s4, s5
remove user	18	Account deletion	T1 s16
rename	19	Change password / username	T1 s16
repeated	35	Duplicate-message detection	Extra s8
reply to a message	6	Replies to a message, threads, quoting	Extra s9
report	3	Reports / moderation (the practice)	T1 s4; T4 s3, s5; T5 s2, s3
reputation	25	Leaderboards / karma	T4 s8
restore	4	Hide / soft delete / restore	T2 s4, s5
results	10	Polls	T1 s10
revert	38	Undo / redo admin actions	Extra s4
revision	5	Edit + edit history (+ undo)	T1 s15
role	16	Moderator role, roles	T1 s16
same message again	35	Duplicate-message detection	Extra s8
save	9	Bookmarks / favourites	T1 s6
saved	9	Bookmarks / favourites	T1 s6
schedule	28	Scheduled / delayed messages	Extra s6
scheduled	28	Scheduled / delayed messages	Extra s6
score	2	Up/down votes, score	T1 s7; T4 s6; T5 s6, s9
scroll	31	Pagination	T2 s9
search	23	Search + autocomplete	Extra s1
section	22	Boards / categories / subforums	Extra s7
seen	27	Read / unread tracking	T1 s11
segment tree	F8	Skip list / heap / hash table / B-tree / Fenwick tree	not coded: T2 s11 (copy RedBlackTree's shape)
session	20	Login lockout + login history	Extra s11
shiftBackward	F9	SortedDataSlice / AVLTreeSlice	not coded: week 6 lab (hidden implementation)
shiftForward	F9	SortedDataSlice / AVLTreeSlice	not coded: week 6 lab (hidden implementation)
shown first	8	Pin messages	T2 s5
since	32	Time-range queries / counts	T2 s9
skip list	F8	Skip list / heap / hash table / B-tree / Fenwick tree	not coded: T2 s11 (copy RedBlackTree's shape)
slice	F9	SortedDataSlice / AVLTreeSlice	not coded: week 6 lab (hidden implementation)
slice of messages	31	Pagination	T2 s9
soft delete	4	Hide / soft delete / restore	T2 s4, s5
SortedArrayList	F3	SortedArrayList backwards + bounds	T2 s7
spam	34	Spam / rate limiting	T1 s17
split	30	Move message / merge / split posts	T2 s12; Extra s12
statistics	26	Statistics	Extra s2
stats	26	Statistics	Extra s2
sticky	8	Pin messages	T2 s5
subforum	22	Boards / categories / subforums	Extra s7
subscribe	12b	Subscribe + notifications	T1 s8
survey	10	Polls	T1 s10
suspend	17	Bans with expiry	T1 s12
suspension	17	Bans with expiry	T1 s12
swear	33	Profanity censor / word filter	Extra s3

Keyword	#	Theme	Code
tag	21	Tags / hashtags	T1 s9
tagged user	7	@mentions	T1 s14
thread	6	Replies to a message, threads, quoting	Extra s9
time window	32	Time-range queries / counts	T2 s9
timeout	17	Bans with expiry	T1 s12
too many	34	Spam / rate limiting	T1 s17
top users	25	Leaderboards / karma	T4 s8
trending	24	Trending / busiest / active posts	T4 s7
undo	38	Undo / redo admin actions	Extra s4
unpinned	F9	SortedDataSlice / AVLTreeSlice	not coded: week 6 lab (hidden implementation)
unread	27	Read / unread tracking	T1 s11
unread notifications	12b	Subscribe + notifications	T1 s8
until	17	Bans with expiry	T1 s12
upvote	2	Up/down votes, score	T1 s7; T4 s6; T5 s6, s9
url	41	Links / emoji / markdown / previews	Extra s8, s13
@username	7	@mentions	T1 s14
view	F7	Filtered view of a SortedData	T2 s3
visibility	4	Hide / soft delete / restore	T2 s4, s5
visible	4	Hide / soft delete / restore	T2 s4, s5
visible to	F7	Filtered view of a SortedData	T2 s3
vote	2	Up/down votes, score	T1 s7; T4 s6; T5 s6, s9
who can view	15	Private / invite-only posts	Extra s10
winner	10	Polls	T1 s10
XML	37	Export / import, JSON	T3 s6, s7

Every predicted theme

Grouped like HACKATHON_PREDICTIONS.md. "Main classes / methods" are the names to search for in the file.

A. Interactions on messages

#	Theme	Code	Main classes / methods
1	Reactions	T1 s5; T4 s4; T5 s5, s8	ReactionTools.addReaction / changeReaction / countReactions / getTopMessages
2	Up/down votes, score	T1 s7; T4 s6; T5 s6, s9	VoteTools.vote / getScore / getTopMessages
3	Reports / moderation (the practice)	T1 s4; T4 s3, s5; T5 s2, s3	ModerationTools.addReport / getReportedMessages; ReportQueries
4	Hide / soft delete / restore	T2 s4, s5	Post.setHidden / setDeleted / getVisibleMessages; ContentTools
5	Edit + edit history (+ undo)	T1 s15	EditTools.editMessage / getHistory / undoLastEdit; Post.replaceMessage
6	Replies to a message, threads, quoting	Extra s9	ThreadTools.reply / getReplies / getThread / quote
7	@mentions	T1 s14	MentionTools.extractUsernames / getMentions
8	Pin messages	T2 s5	ContentTools.setPinned / getMessagesPinnedFirst
9	Bookmarks / favourites	T1 s6	SocialTools.bookmark / getBookmarks
10	Polls	T1 s10	PollTools.createPoll / vote / changeVote / getResults / getWinners

B. Users and access

#	Theme	Code	Main classes / methods
11	Block / mute users	T1 s6	SocialTools.block / isBlocked / getMessagesFor
12	Follow users	T1 s6; T5 s7	SocialTools.follow / getFollowers
12b	Subscribe + notifications	T1 s8	NotificationTools.subscribe / getNotifications / markAllRead
13	Friend requests	T1 s13	FriendTools.sendRequest / accept / areFriends
14	Direct / private messages	Extra s5	DirectMessageTools.send / getConversation / getLatest
15	Private / invite-only posts	Extra s10	AccessTools.setPrivate / invite / canView
16	Moderator role, roles	T1 s16	Permissions.setModerator / canModerate; AccountTools.setRole
17	Bans with expiry	T1 s12	BanTools.ban / unban / isBanned
18	Account deletion	T1 s16	AccountTools.deleteAccount / displayName
19	Change password / username	T1 s16	AccountTools.changePassword / changeUsername
20	Login lockout + login history	Extra s11	LoginTools.login / isLocked / getHistory / unlock

C. Organising content

#	Theme	Code	Main classes / methods
21	Tags / hashtags	T1 s9	TagTools.addTag / getPostsWithTag / getPopularTags
22	Boards / categories / subforums	Extra s7	Board (Composite), BoardTools
23	Search + autocomplete	Extra s1	SearchTools.search / autocompleteUsername; Trie
24	Trending / busiest / active posts	T4 s7	TrendingTools.getTrendingPosts / getTopPosts
25	Leaderboards / karma	T4 s8	LeaderboardTools.getTopUsers
26	Statistics	Extra s2	StatsTools
27	Read / unread tracking	T1 s11	ReadTracker.markRead / getUnreadCount
28	Scheduled / delayed messages	Extra s6	ScheduledMessages.schedule / publishDue / cancel
29	Lock / close a post	T2 s5, s6	ContentTools.setLocked / reply; MemberState.canReplyTo
30	Move message / merge / split posts	T2 s12; Extra s12	MoveTools.moveMessage; PostTools.mergePosts / splitPost
31	Pagination	T2 s9	MessageQueries.getPage
32	Time-range queries / counts	T2 s9	MessageQueries.getMessagesBetween / countMessagesBetween

D. Content safety

#	Theme	Code	Main classes / methods
33	Profanity censor / word filter	Extra s3	Censor, CensorFactory, CensorStrategy
34	Spam / rate limiting	T1 s17	RateLimiter.tryAcquire
35	Duplicate-message detection	Extra s8	ContentChecks.isDuplicate

E. Infrastructure

#	Theme	Code	Main classes / methods
36	Notifications inbox	T1 s8	NotificationTools
37	Export / import, JSON	T3 s6, s7	JSONFormattedFactory / JSONWriter / JSONReader; Features
38	Undo / redo admin actions	Extra s4	AdminActions.run / undo; AdminCommand
39	O(1) lookup index	T1 s3	MessageIndex; UserDAO.getByUUID (indexed)
40	Audit log	Extra s4	AdminActions.getLog; AuditEntry
41	Links / emoji / markdown / previews	Extra s8, s13	ContentChecks.extractLinks; TextFormat

F. Trees and data structures

#	Theme	Code	Main classes / methods
F1	AVL delete	T2 s7	AVLTree.remove; AVLNodeFilled.delete / rebalance
F2	BSTree.getRange (was null)	T2 s10	BSIterator
F3	SortedList backwards + bounds	T2 s7	SortedList.getRange(..., true) / getAtIndex
F4	Red-black tree	T2 s11	RedBlackTree (left-leaning)
F5	Order statistics	T2 s8, s9	rank / floor / ceiling; MessageQueries.getPosition
F6	floor / ceiling / successor / predecessor	T2 s8	SortedList.floor / ceiling
F7	Filtered view of a SortedData	T2 s3	SortedListViews.filter; Post visible tree
F8	Skip list / heap / hash table / B-tree / Fenwick tree	not coded: T2 s11 (copy RedBlackTree's shape)	not built: only if the spec names one. RedBlackTree shows every SortedData method to implement
F9	SortedListSlice / AVLTreeSlice	not coded: week 6 lab (hidden implementation)	not built: the course hides it; reuse your week 6 AVLSliceTests

Spec sentence -> what to write

If the question contains one of these sentences, this is the tested way to handle it.

The spec says	Do this
"if X or Y does not exist, do nothing and return false"	MessageIndex.get / UserDAO.getByUUID, both O(1). T1 s3
"already reported / reacted ... return false"	map.putIfAbsent(user, x) != null -> false. T1 s4
"return false if the user was not reporting"	map.remove(user) == null -> false. T1 s4
"performance is important"	no scans inside add / remove / has; HashMap + indexes; say it in a comment
"Admin users should be able to ..."	UserDAO.getByUUID(user).role() == User.Role.Admin. T2 s4
"visible by default" / "non-admins only see ..."	second tree kept in sync, or SortedDataViews.filter. T2 s3
"strategy is either ... throw an exception otherwise"	Factory switch, default -> IllegalArgumentException; also null. T4 s3
"amount ... fewer if insufficient ... at most once ... no zero"	TopKIterator: include filter + amount + one group per message. T4 s4
"ties may be returned in arbitrary order"	do not tie-break; tests compare as sets. T5 s3
"persist across runs ... other languages ... portable"	CSV one row per fact (or JSON); fix CSVReader.hasNext + IO date check first. T3 s2, s6
"must use the Iterator and Factory patterns"	ReportOrderingFactory + ReportedMessageIterator / TopKIterator. T4 s3
"branch-complete coverage on X including submethods"	branch map table in javadoc, one test per outcome. T5 s2
"distinguish between a correct and faulty implementation"	assert only specified behaviour; make bugs change the ORDER. T5 s3, s9

Traps in the base code (each one cost marks in the practice base)

Trap	Fix
No message lookup by id	Messages sit in each post's tree sorted by (time, thread, poster, id). Use MessageIndex (T1 s3.2).
UserDAO.getByUUID is O(n)	Index users by UUID; the map is created in clear() because DAO's constructor calls clear() first (T1 s3.1).
Null poster / thread in tests	MessageComparator and the serializers crash on null: null-safe versions in T1 s3.3 and T3 s4.
CSVReader.hasNext always true	Reading any file ends in an exception. Peek with a PushbackReader (T3 s2.1).
ComputerIOFactory date check	Every read returns null after Dec 2025, so nothing loads. Remove it, create the folder (T3 s2.2).
Message / User are records	Immutable: keep new state in a store keyed by UUID, or replace the record keeping the UUID (T2 s4, T1 s16).
Editing a message	The comparator ignores the text: remove the old record BEFORE inserting the new one (T2 s4 replaceMessage).
AVL balance sign	Practice base: left - right. Your GitLab main: right - left. Flip the comparisons to match (T2 s7).
Static singletons in tests	Clear every DAO and store in @Before, or tests depend on run order (T5 checklist).
valueOf(null) / switch on null	Throws NullPointerException; specs want IllegalArgumentException. Check null first (T4 s9).
Black-box tests	Never assert tie order or unspecified behaviour; design data so a bug changes the ORDER (T5 s3, s9).
Must compile	A team commit that does not compile scores zero for code. Compile before every push to main.

Patterns -> where they are used

Pattern	In the code
Iterator	ReportedMessageIterator, TopKIterator, BSIterator (T4 s3-s4, T2 s10)
Factory	ReportOrderingFactory, ReactionOrderingFactory, CensorFactory, VoteTools.ordering
Strategy	the Comparators the factories return; CensorStrategy; PersistentFeature
Singleton	DAOs, MessageIndex, ReportStore, ReactionStore
Observer	SortedData listeners -> MessageIndex -> notifications / mentions / search
Facade	every *Tools class
State + Template Method	MemberState.canReplyTo overridden by AdminState (locked posts)
Command	AdminCommand + AdminActions undo
Memento	Edit history
Composite	Board tree
Decorator	chained Censor